Machine Control

GNSS technology is being integrated into equipment such as bulldozers, excavators, graders, pavers and farm machinery to enhance productivity in the real-time operation of this equipment, and to provide situational awareness information to the equipment operator.



Precision Agriculture

In precision agriculture, GNSS-based applications are used to support farm planning, field mapping, soil sampling, tractor guidance, and crop assessment. More precise application of fertilizers, pesticides and herbicides reduces cost and environmental impact.

Surface Mining

GNSS information is being used to efficiently manage the mining of an ore body and the movement of waste material. GNSS equipment installed on shovels and haul trucks provides position information to a computer-controlled dispatch system to optimally route haul trucks to and from each shovel.

Survey

Using GNSS, it is possible for a single surveyor to accomplish in one day what might have taken a survey crew of three people a week to complete. Determining a new survey position once required measuring distances and bearings from an existing (known) survey point to the new point.



GLOSSARY

1. **Microprocessors** - an integrated circuit that contains all the functions of a central processing unit of a computer.
2. **Transponder** - a device for receiving a radio signal and automatically transmitting a different signal.
3. **Lithological** – relating to the general physical characteristics of rocks.
4. Artificial Intelligence - development of computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.
5. **Digitization** - the conversion of text, pictures, or sound into a digital form that can be processed by a computer.
6. **Geosynchronous** - it is an orbit around Earth of a satellite with an orbital peroid that matches Earth's rotation on its axis.
7. **Radar** - a system for detecting the presence, direction, distance, and speed of aircraft, ships, and other objects.
8. **Photogrammetry** - the use of photography in surveying and mapping to ascertain measurements between objects.
9. **Automation** - The use of automatic equipment in a manufacturing or other process.
10. **Herbicide** - a substance that is toxic to plants, used to destroy unwanted vegetation.

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